Expand and Exempt: Compensation and the Politics of US Chip Export Controls*

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Abstract

Since October 2022 the United States has widened its export controls on advanced chips and chipmaking equipment, although nearly every commenting firm and trade association asked for less. I ask whether the controls are security statecraft or private commercial interest presented in the language of security. The evidence comes from public comments on 49 federal dockets, scored with a fine-tuned language model validated on 213 hand-coded letters, seven rule revisions and their 58 carve-outs, lobbying disclosures and CHIPS Act awards. The controls are security statecraft that private interest hollows out from within. Chipmakers and their suppliers of tools and materials asked for narrower controls than firms that buy and deploy chips. The Bureau of Industry and Security widened scope in five of six chip-rule revisions over that opposition. Yet those rules reopened flows into China for the firms that resisted most. Chipmaking and equipment firms received 18 of the 24 carve-outs that did so, including all 8 that BIS granted at commenters' request, although their products are the chokepoints behind US leverage over Chinese chipmaking. They also received $32.8 billion of $33.5 billion in CHIPS awards, and firms whose awards were finalized lobbied less afterward than firms still waiting.

*Working paper in progress. Please do not cite or circulate.

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1 Introduction

On October 7, 2022, the Bureau of Industry and Security (BIS) at the US Department of Commerce placed license requirements on advanced computing chips, on the computers that contain them and on equipment used to make advanced chips, and it restricted what US persons could do for Chinese fabrication plants. By April 2026 BIS had revised its chip rules six times. Each revision followed a public comment period, and in each period the firms and trade associations that wrote in asked BIS to control less. BIS widened scope in five of the six revisions anyway. Yet the same revisions also reopened flows into China, and those exemptions went mostly to the firms that had resisted hardest, the makers of the equipment and materials that give the United States its hold over Chinese chipmaking.

The sequence poses a question that the debate over US technology policy toward China has not settled. Are chip export controls economic statecraft driven by national security, or are they private commercial interests presented in the language of national security? The security reading has strong support. Advanced chips train frontier AI models, the United States and its allies hold chokepoints in chip design and equipment (Buchanan, 2020; Farrell & Newman, 2019; Sastry et al., 2024), and the October 2022 rule states that BIS sought to restrict China's "access to advanced computing for its military modernization, including nuclear weapons development" (87 FR 62186). The commercial reading also has support. Security exceptions have long sheltered economic aims in trade law (Heath, 2020; Pinchis-Paulsen, 2020), export controls historically bent to exporters (Mastanduno, 1992),

and the controls arrived alongside $39 billion in manufacturing subsidies under the CHIPS and Science Act.

I argue that chip export controls are security statecraft that private interest hollows out from within. The state set the direction of the controls, and the firms with the most to lose decided where they would bite. The link between the two is compensation. Controls on chips and chipmaking equipment impose their heaviest costs on firms that make chips and the tools and materials to make them. Those firms form the base of the AI compute stack, the layers of hardware, cloud computing, models and applications on which AI is built. A state that wants to keep widening such controls has to manage those firms. It can write carve-outs into the rules that soften the controls where those firms are exposed, and it can pay them through industrial policy. Both forms of payment let the state keep expanding scope without the losers turning the rulemaking against it. The carve-outs carry a security cost. The firms that resist most are the ones that hold the chokepoints, so the relief they win reopens part of what the controls were written to close.

The evidence comes from the documentary record of US chip policy between 2022 and 2026. I collect every public comment on 49 federal dockets that bear on export controls, CHIPS incentives and AI policy, 6,641 comments in all, and score each one for whether it asks for wider or narrower control, using a fine-tuned natural language inference model validated out of fold against 213 hand-coded letters. Each commenting organization is placed in a layer of the AI compute stack according to what it does. For the seven BIS revisions that followed a commented chip rule, I code whether scope widened, and for the 58 carve-outs those revisions added, who benefits, which commenting firms each one relieves and whether

BIS says commenters asked for them. Federal lobbying disclosures are joined to the dates of CHIPS preliminary terms and final awards.

I report three findings. First, firms in fabrication, equipment and materials asked for narrower controls than firms that buy and deploy chips and AI. The gap holds with agency or docket fixed effects and with argument type held fixed. Second, BIS widened scope over unanimous opposition in the comment record, and every revision also added carve-outs. Of the 24 carve-outs that reopened flows into China, 18 reach fabrication firms, the coalition that resisted most and whose products give the controls their force. All 8 of the China-flow carve-outs that BIS granted at commenters' request are among them. Items from the fabrication layer make up only about a third of the controlled provisions in those rules, and across all 58 carve-outs, 35 reach fabrication firms. Among the firms that commented, those that pushed harder for narrowing received more carve-outs that relieved them directly. The pattern also holds across 119 firms in the wider corpus, including among firms in the same layer of the stack. Third, CHIPS money went to the same coalition, $32.8 billion of $33.5 billion. Lobbying intensity predicts who received preliminary terms only weakly, and firms whose awards were finalized cut their chip and export-control lobbying relative to firms that received preliminary terms and never finalized. The late expansions against equipment makers and foundries, in December 2024 and January 2025, came in the weeks when most of those firms' awards were finalized.

The paper contributes to three literatures. For work on firms in trade politics (I. S. Kim, 2017; I. S. Kim & Osgood, 2019; Osgood, 2017), it shows the firm-level cleavage carrying over from tariffs to security controls, with position in the AI compute stack sorting firms

within a single industry. For work on exceptions and escape clauses (Pelc, 2016; Rosendorff & Milner, 2001) and on business influence in rulemaking (Libgober, 2020a; Yackee & Yackee, 2006), it shows carve-outs working as the price of expansion, landing on the coalition that resisted most and reopening flows to the country the controls were aimed at. For work on economic statecraft and the new industrial policy (Baldwin, 1985; Bown & Wang, 2024; Farrell & Newman, 2019; Juhász & Lane, 2024; Weymouth, 2025), it ties subsidies and controls together as one bargain, in which the state paid the losers of a security policy and gave back part of its reach.

Section 2 develops the argument and the hypotheses. Section 3 describes the rules, the awards and the data. Sections 4 through 6 report the tests. Section 7 returns to the central question, and Section 8 concludes with what the record cannot show.

2 Theory

Economic statecraft is the use of economic means for foreign policy ends (Baldwin, 1985). On the security reading, chip controls are statecraft of a familiar kind. The United States and a few allies dominate chip design software, advanced equipment and the most capable chips, and a state that controls the hubs of a network can deny them to rivals (Beaumier & Cartwright, 2024; Farrell & Newman, 2019). Compute is one of the inputs that drive AI capability, which makes advanced chips a natural target (Buchanan, 2020; Sastry et al., 2024), and by 2025 the United States held about three quarters of the world's AI compute capacity (Abecassis et al., 2025). Rising powers that threaten an incumbent

invite it to weaponize supply chains (Chen & Evers, 2023), and China's industrial policy in semiconductors posed exactly that threat to US structural power (Malkin, 2022).

The commercial reading starts from the observation that security and economic aims are hard to separate in trade and investment policy (Mastanduno, 1998; Roberts et al., 2019). National security claims have expanded and resist review (Heath, 2020), the GATT security exception was drafted with economic aims in view (Pinchis-Paulsen, 2020), and security screening of foreign investment in Europe tracks national commercial stakes (Chan & Meunier, 2022). Public choice accounts treat sanctions and controls as policies that interest groups shape like any other (Kaempfer & Lowenberg, 1999), and the history of CoCom shows export controls bargained among security agencies, allies and exporters (Mastanduno, 1992). Regulation in general can be acquired by the industry it governs (Stigler, 1971), although claims of capture need evidence that the regulated industry set the content of the rule (Carpenter & Moss, 2013). Securitization theory adds that calling an issue a security matter lifts it out of normal politics (Buzan et al., 1998), which would let commercial rivalry with Chinese firms travel under a security label.

The commercial reading needs US firms that gain from wider controls, and some parts of the industry qualify. Cloud providers and AI developers buy chips, sell little hardware into China and compete with Chinese AI firms. Controls that keep advanced compute from those rivals raise their costs, and control over rivals is one of the main things an industry seeks from regulation (Stigler, 1971). A security label makes such a request easier to defend. US chipmakers that compete with Chinese producers gain in the same way when controls slow those producers down. Firms that want wider controls have several ways into

the rules. They can ask for wider scope in public comments (Yackee & Yackee, 2006), press their case in meetings with officials (Libgober, 2020a) and shape drafts before publication, since agencies write proposals with the expected comments in mind (Libgober, 2020b). On this reading, each revision should move toward the coalition that asked for it, and relief should go to the firms with the most influence.

The security reading predicts something different. If security set the controls, the direction should hold against business opposition. Both readings expect the losers to be managed, since a state that pursues a security aim at the expense of domestic firms still has to deal with those firms. They differ on what the relief gives up. On the security reading, carve-outs should trim the controls' reach into domestic and allied operations and leave the flows to the target closed. On the commercial reading, carve-outs should follow the firms with the most at stake, even where they reopen those flows.

2.1 Who Loses from Chip Controls

Firm-level trade politics holds that firms within one industry want different trade policies. Large, productive firms in differentiated industries lobby on their own and for their own products (Bombardini & Trebbi, 2012; I. S. Kim, 2017). Product variety and intra-industry trade break the unified industry opposition that sector models assume (Madeira, 2016; Osgood, 2017). Firms embedded in global supply chains support openness and back trade agreements (Milner, 1988; Osgood, 2018; Zeng et al., 2020), and they hold preferences over several dimensions of trade policy at once (I. S. Kim et al., 2019).

Export controls bring this logic into security policy. Studies of firms in trade usually place them in a value chain, a sequence of stages in which each stage supplies the next (Gereffi et al., 2005), and production-network research has carried that lens to AI (Ferrari, 2023). Recent work describes the industry that chip controls fall on as a stack, a term Bratton (2016) brought to the study of computation and sovereignty. Its layers run from the hardware that performs computation through cloud computing and models to the applications built on them (Hagiu & Wright, 2025; Jacobides et al., 2021; Narechania & Sitaraman, 2024; van der Vlist et al., 2024). Each layer is the platform for the one above, and a few firms hold each layer. One firm designs most data center GPUs, a few providers dominate cloud computing, and model developers are tied to those providers by investments and exclusive contracts (Korinek & Vipra, 2025; Luitse, 2024; Sastry et al., 2024; Widder et al., 2024), so the firms on either side of each link depend on one another.

The same concentration gives states leverage. Narechania and Sitaraman (2024) see national security risks at the critical points of the stack. Hyperscale cloud firms have become the hubs through which states can weaponize interdependence (Lehdonvirta et al., 2025), and states that host computing capacity can govern AI through their territory (Lehdonvirta et al., 2024). States now use trade policy to secure control over the AI stack, and their strategies depend in part on where they sit in it (Weymouth, 2025). The US government adopted the stack as its own frame in July 2025. An executive order directed Commerce to promote exports of a "full-stack AI technology package" that runs from chips, servers and cloud services through data pipelines and models to applications (90 FR 35393).

I divide the stack into five layers (Figure 1). Fabrication is the base. It holds the

makers of equipment and materials, the fabs and the firms that package and test chips, and every layer above depends on what they can make. The design layer holds chip designers, licensors of intellectual property, makers of design software and the research they draw on. Compute, models and applications sit above design, and I group them as the deployment layers. In value-chain terms the fabrication layer is the midstream, the design layer is the upstream and the deployment layers are the downstream. I call the firms in these layers fabrication firms, designers and deployers. A value chain orders its stages by the sequence of production, while the stack orders them by dependence, which places fabrication beneath design. The same rule falls on each layer differently.

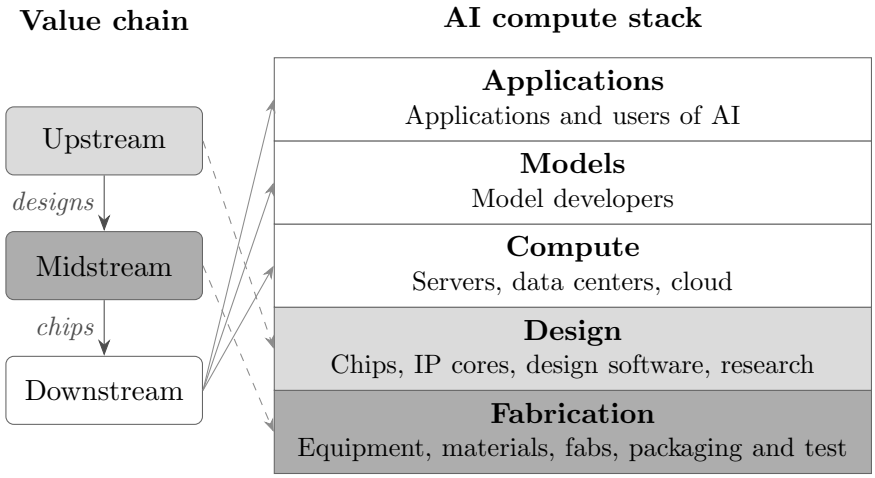


Figure 1. Stages of the value chain and layers of the AI compute stack. Each layer of the stack rests on the one below it.

Controls on chipmaking equipment and on fabrication hit fabrication firms hardest, and controls on chips reach them as well, since foundries and packagers make the controlled chips. The controlled items are their products. China is the largest market for the major US equipment makers, with 37 percent of Applied Materials' revenue, 42 percent of Lam Research's and 43 percent of KLA's in fiscal year 2024 (Applied Materials, Inc., 2024; KLA

Corporation, 2024; Lam Research Corporation, 2024), and export controls on Chinese customers impose large losses on their US suppliers (Crosignani et al., 2026). Equipment controls also reach fabrication firms' own operations. The October 2022 rule required a license for US persons to service tools in Chinese fabs that make advanced chips (§ 744.6(c)(2)(iii), 87 FR 62199) and for any item subject to the Export Administration Regulations (EAR) that would be used to develop or produce chipmaking equipment or its parts in China (§ 744.23(a)(2)(v), 87 FR 62201). Fabs and memory makers with plants in China face license requirements on their own facilities (§ 744.23(a)(2)(iii), 87 FR 62201). From October 2023, Samsung's and SK hynix's memory plants in China received controlled items under general Validated End-User authorizations (88 FR 71479), which BIS revoked effective December 31, 2025 (90 FR 42321).

Designers lose sales of controlled chips, and those losses can be large. Their exposure, though, depends on where the rule draws its thresholds, and they can redesign products to fall below them. Designers also have large buyers at home. Cloud computing and model development are concentrated in a few firms, many of them based in the United States (Sastry et al., 2024; Weymouth, 2025), and their demand for chips can absorb part of what designers lose in China. Deployers lose least. They buy chips, and controls on sales to China do not raise their input costs at home. Cloud providers and AI developers among them can even gain from controls on their rivals' inputs.

Exposure also varies among firms in the same layer. A rule reaches a firm through each controlled product it sells and each plant, customer or supplier it has in China, so a toolmaker with many product lines meets the controls at more points than a specialist does.

The firms with the most at stake have the strongest reason to press their own case, and the largest of them can press it alone (Bombardini & Trebbi, 2012; I. S. Kim, 2017).

Opposition should therefore fall as one moves up the stack. Fabrication firms should oppose expansion most, designers should oppose it where thresholds catch their products, and deployers should oppose it least. Designers' exposure turns on thresholds that change from rule to rule, so the stable contrast runs between the fabrication layer and the deployment layers.

H1. Firms in the fabrication layer oppose control expansion more strongly than firms in the deployment layers.

2.2 How the State Pays and What Payment Buys

A state that widens controls over the objections of fabrication firms can pay them in two ways. The first is regulatory. Rules can be written with exceptions, exclusions and transition periods that soften the control where specific firms are exposed. Trade agreements include escape clauses because flexibility lets governments absorb domestic pressure while keeping the larger commitment (Kucik & Reinhardt, 2008; Rosendorff & Milner, 2001), and how exceptions are designed and who can invoke them is a political choice (Pelc, 2016). Regulatory barriers tend to favor firms large enough to absorb their costs and shape their terms (Gulotty, 2020). Firm-specific relief processes in trade, such as tariff suspensions and exclusions, respond to the information and political connections of the firms that request them (Fotak et al., 2025; Ludema et al., 2018). In the exclusion process for the tariffs on

Chinese goods, lobbying helped firms win the exact relief they requested, and it worked far better for large firms than for small ones (Eldes et al., 2026). Commerce's own exclusion process for the Section 232 steel tariffs approved requests in line with political alignment and trade ties with the exporting country, and requests from firms that lobbied were less likely to succeed (Born et al., 2024). In export controls the equivalent instruments are license exceptions, exclusion notes on ECCNs, temporary general licenses, savings clauses and routes by which approved parties can certify their way out of a control. Each is written into the Federal Register text, and each names whom it relieves. In an exclusion process a firm asks for relief on a named product and the agency grants or denies the request. A carve-out is a general provision that the agency drafts, and the agency decides which exposures it covers and in what form.

The structure of the stack makes this targeting practical. A few firms hold each layer (Narechania & Sitaraman, 2024; Sastry et al., 2024), so a provision drafted around one exposure, such as servicing tools in Chinese fabs or supplying memory plants in China, reaches a small set of firms that the agency can name in advance. It can write relief in general terms and still know whom the relief will reach.

Notice-and-comment rulemaking is where firms ask for these instruments. Business comments move final rules more than other comments (Yackee & Yackee, 2006); firms that meet with regulators gain distributive benefits (Libgober, 2020a); larger and more diverse coalitions move rules further (Dwidar, 2022; Nelson & Yackee, 2012); and firms keep lobbying after a policy is enacted to shape how it is implemented (You, 2017). Agencies also anticipate comments when they draft, so observed changes understate influence on the draft itself

(Libgober, 2020b); and comments that appear independent can carry their funders' interests (Bertrand et al., 2021).

Accounts of rulemaking differ on how relief should track requests. If the agency answers the comments in front of it, as the success of business commenters suggests (Yackee & Yackee, 2006), relief should follow what each firm asked for. If the state is managing resistance, it chooses the form of payment itself. Relief should then follow how hard a firm resists and how much it stands to lose, whether or not the firm asked for the provision that relieves it. Firms also reach the agency through channels the comment docket does not record. Agencies draft with expected comments in mind (Libgober, 2020b), firms meet with officials (Libgober, 2020a), and lobbying during White House review of draft rules is associated with changes to them, most of all when only business groups lobby (Haeder & Yackee, 2015).

If carve-outs are part of the price of expansion, they should arrive with it, and they should go to the firms whose resistance the state most needs to manage.

H2a. BIS revisions that widen control scope also add carve-outs.

H2b. Carve-outs go disproportionately to the coalition that opposes expansion most.

The second way is fiscal. Compensation has long sustained support for costly economic policy. Embedded liberalism paired openness with domestic insurance (Ruggie, 1982), open economies run larger governments (Rodrik, 1998), and compensation sustains public

support for openness (Hays, 2009; Walter, 2010). Adjustment assistance and protection work as substitutes (S. E. Kim & Pelc, 2021), and assistance to those hurt by trade dampens backlash (Ritchie & You, 2021). Issue linkage builds domestic support for costly agreements (Davis, 2004). In climate policy, industrial policy builds coalitions that later support costly regulation (Meckling et al., 2015), and targeted compensation raises support for transition among those it pays (Bolet et al., 2024; Gaikwad et al., 2022). The return of industrial policy (Evenett et al., 2024; Juhász et al., 2024) gives states a new set of instruments for this purpose, and chip subsidies, tariffs and controls form one policy package (Bown & Wang, 2024; Luo & Van Assche, 2023). Industrial policy also raises the question of who captures it (Breznitz & Gingrich, 2025; Juhász & Lane, 2024).

Fiscal compensation is competed for. The CHIPS Act left the allocation of its manufacturing incentives to the Department of Commerce, the department in which BIS also writes the export controls. Firms applied under a notice of funding opportunity. Commerce signed a preliminary memorandum of terms with applicants it judged "reasonably likely to receive an award," reviewed their projects and then negotiated final terms (Congressional Research Service, 2026). At each step Commerce decided who would be paid, how much and on what conditions. That kind of discretion leaves room for lobbying to shape the outcome. When firms applied for capital under the Troubled Asset Relief Program, those with political connections, including lobbying and campaign contributions, were more likely to be funded (Duchin & Sosyura, 2012).

The firms that fight the controls are also well placed to win this competition. Lobbying on export controls shows how much a firm stands to lose, and a state that pays

losers to keep a policy going has reason to direct its money toward them. The same lobbying gives a firm working access to the department that decides its award. Access brings firms distributive benefits in rulemaking (Libgober, 2020a), and a firm already in regular contact with Commerce over the scope of the controls can carry that relationship into the negotiation of its award. The payment is therefore partly a product of the firm's political activity, which makes the step from lobbying to award a separate question from the step from award to behavior. Firms that lobby harder through the application process should be more likely to come away with an award.

H3a. Firms that lobby more heavily on export controls are more likely to secure a CHIPS award.

What payment changes depends on the terms the firm signs. A preliminary memorandum of terms does not guarantee an award (Congressional Research Service, 2026). A firm that holds one still has money to win, and its lobbying on chips and export controls serves its case for the award as well as its case against the controls. Finalization turns the terms into a contract that binds the firm to the national security guardrails of the CHIPS Act. For ten years after the award, a recipient that materially expands semiconductor manufacturing capacity in China or another foreign country of concern can be made to repay the full amount, with exceptions for legacy chips (88 FR 65600, 65601). A firm that has accepted that limit has less at stake in the Chinese plants that the controls also reach. Commerce pays out awards as firms complete negotiated project milestones, and some projects are scheduled to run into the 2030s (Congressional Research Service, 2026), so a paid firm

depends on the department for years after its award is signed.

Payment acts mainly on how hard a firm fights. A paid firm can still believe the controls cost it money. A firm that has received relief in the rule text has less left to ask for, though, and once its award is secured it has less to gain from pressing its case. Because its disbursements run through Commerce, it also has more to lose from a public fight with the department. Payment should therefore reduce a firm's mobilization against the controls, whatever happens to its stated position.

H3b. Firms whose CHIPS award finalizes reduce their export-control lobbying relative to firms that reached preliminary terms and never finalized.

If the firms that resisted were paid and their resistance faded, the controls could keep widening without the losers setting their terms, and the state would keep the direction of policy while conceding its distribution.

3 Data and Measurement

3.1 The Comment Corpus

I collected every public comment on 49 dockets from Regulations.gov, 6,641 comments filed between 2022 and 2026 to 16 agencies. The core is the eight BIS dockets on the chip, advanced computing and AI rules described in Section 3.2 and the Commerce Department's 2022 request for information on CHIPS incentives (DOC-2022-0001). The

rest are other BIS dockets on export controls, on supply-chain rules for connected vehicles and drones and on Section 232 investigations of critical minerals and robotics, and every non-BIS docket since 2022 with artificial intelligence in a document or docket title and at least one public comment, among them dockets at the National Telecommunications and Information Administration (NTIA), the Office of Science and Technology Policy (OSTP), the National Institute of Standards and Technology (NIST), the Office of Management and Budget (OMB), the Federal Trade Commission (FTC), the International Trade Administration (ITA) and the Patent and Trademark Office. Many firms file comments as PDF attachments, so the text of each comment includes its attachments, and letters that BIS posted as page images were recovered by optical character recognition. Comments under 500 characters (762) cannot state a position on scope and are dropped, which leaves 5,879 scored comments.

Campaign copies and coordinated letters are common. On the 2024 round of the advanced computing rule, letters from three industry groups share hundreds of identical twelve-word passages, including drafted regulatory text, and on the equipment docket Applied Materials, KLA and Lam Research filed recommendations from one template. I group comments into clusters that share at least half of their sampled twelve-word strings or come from the same organization, 5,570 clusters in all, and cluster standard errors on them.

3.2 Rules and Awards, 2022 to 2026

Table 1 lists the BIS rules whose dockets generate the corpus. The October 2022 interim final rule did four things. It created license requirements for advanced computing chips (ECCN 3A090) and computers containing them (4A090) destined for China; controlled equipment for producing advanced-node chips; extended end-use controls to supercomputer and semiconductor manufacturing in China; and required licenses for US persons supporting advanced-node production there. BIS issued it as an interim final rule, effective before comment, and invited comments afterward.

Table 1. BIS semiconductor and advanced computing rules with a public comment record, 2022 to 2025

Effective	FR document	Citation	Rule	Comments
2022-10-07	2022-21658	87 FR 62186	Advanced computing and SME controls	49
2023-11-17	2023-23049	88 FR 73424	SME rule	23
2023-11-17	2023-23055	88 FR 73458	Advanced computing update	41
2024-03-13	2024-05267	89 FR 18353	Radiation-hardened ICs, License Exception GOV	5
2024-04-04	2024-07004	89 FR 23876	Corrections and clarifications	15
2024-12-02	2024-28270	89 FR 96790	FDP additions, HBM and SME refinements	13
2025-01-13	2025-00636	90 FR 4544	AI diffusion framework (not enforced)	67
2025-01-16	2025-00711	90 FR 5298	Foundry due diligence	15
2025-09-29	2025-19001	90 FR 47201	Affiliates rule	29
2025-11-10	2025-19846	90 FR 50857	One-year suspension of the affiliates rule	0

Note. Effective dates and comment counts from the Federal Register API. SME, semiconductor manufacturing equipment. FDP, foreign direct product rule. HBM, high bandwidth memory. Uncommented corrections (2023-27588, 2025-02655) and the April 2026 designer-date extension (2026-06851) omitted.

Revisions came in waves. In October 2023 BIS split the rule in two, one on advanced computing and one on equipment, each responding at length to the comments received. The computing rule redefined the chip thresholds to capture chips designed to fall just below the 2022 line and added a notification-based license exception. The equipment rule folded the 2022 equipment controls into existing ECCNs, extended license requirements to

all arms-embargoed destinations (Country Group D:5) and narrowed several provisions that commenters had called overbroad. A corrections rule followed in April 2024. In December 2024 BIS added new controls on equipment, two foreign direct product rules that reach equipment made abroad with US technology, and controls on high bandwidth memory. In January 2025 it issued the AI diffusion framework, which Commerce announced on May 13, 2025, two days before its compliance date, that it would not enforce and would rescind (U.S. Government Accountability Office, 2026), and a foundry due diligence rule that requires foundries and packaging firms to presume that certain advanced chips are controlled unless the designer or packager is on an approved list. In September 2025 BIS extended Entity List restrictions to any firm at least 50 percent owned by a listed entity. Six weeks later it suspended that affiliates rule for a year.

The CHIPS and Science Act, signed on August 9, 2022, appropriated $39 billion for manufacturing incentives through fiscal year 2026 (Congressional Research Service, 2026). The CHIPS Program Office announced preliminary memoranda of terms from December 2023 through January 2025 and finalized most awards between November 2024 and January 2025. Twelve firms received preliminary terms and no final award. Figure 4 plots the dates.

After January 2025 the channel changed. The White House announced the one-year suspension of the affiliates rule on November 1, 2025, as part of a trade arrangement with China (The White House, 2025), and BIS implemented it on November 10 by a final rule without notice and comment.[1] In January 2026 BIS moved license review for Nvidia's H200 and less advanced chips to China from a presumption of denial to case-by-case review,

[1]The suspension rule itself (90 FR 50857) gives no reason beyond continued evaluation of national security and foreign policy interests.

again by a final rule issued without prior comment (91 FR 1684). Decisions on chip sales moved from the rulemaking record to the White House. Section 7 returns to this shift.

3.3 Stance on Control Scope

The outcome for H1 is each comment's stance on control scope, from −2 (asks for permanent narrowing, such as an exclusion, a raised threshold or withdrawal) through 0 (no net position) to +2 (asks that more be controlled). On the CHIPS docket the same scale measures stance on the conditions attached to CHIPS money. On the other dockets it measures whether the commenter wants the rule or the policy to control more or less.

I hand coded 213 letters, the 182 eligible comments on the eight BIS chip dockets and a stratified sample of 31 from the CHIPS docket, under a written codebook (Appendix A.4). Each letter is coded on its own text, so a filer's other letters do not carry a code across. The hand codes serve as training data and validation, and every test uses model scores.

The scorer is a DeBERTa-v3 natural language inference model (Laurer et al., 2024) fine-tuned on the hand codes. It judges each comment against four statements (the commenter wants the rule to control less, supports it or wants more, asks for specific narrowing, asks for specific widening) whose probabilities combine into the five-point scale. A second head assigns one of five argument types, which record where the commenter argues from (design inputs, fabrication, deployment, compliance process, or national security). Training is cross-fitted in five folds, with folds grouped by template cluster so that copies of one letter

never sit on both sides of a split, and gold comments are scored by the model that did not see them. The model reads at most 512 tokens at a time, which covers about the first third of the median comment. I therefore split each comment into windows of 1,800 characters, score every window and combine the window scores into one score per comment. Encoders with longer inputs can read most comments whole. Refit with ModernBERT, which reads up to 8,192 tokens (Warner et al., 2025), on the same hand codes and folds, the scorer agrees less with the hand codes than the windowed version (Appendix A.2).

Out of fold, the scorer's stance correlates with the hand codes at 0.78 (Spearman) across all 213 letters and at 0.73 on the BIS letters, against 0.64 for the same model before fine-tuning (Table A1). Argument type matches the hand code for 78 percent of letters. Validation is weaker on the CHIPS docket, where the correlation is 0.49, and absent on the AI dockets, which have no hand codes. I therefore read the pooled results through specifications that compare commenters within agency or within docket. Table 2 shows five letters with their hand codes and out-of-fold model scores.

3.4 Position in the Stack

H1 needs each commenter's layer in the AI compute stack. I code it from what the organization does, the same in every docket. The fabrication layer covers equipment, materials, fabrication, packaging and test. The design layer covers chip design, IP cores, design software and research. The deployment layers cover everyone who buys or uses chips or AI, and the coding does not separate compute, models and applications. These codes

Table 2. Five comment letters, hand codes and model scores

Commenter	Layer	Excerpt	Hand code	Model score
KLA BIS-2023-0016	Fabrication	"Several of the comments are focused on ensuring a level regulatory playing field between U.S. semiconductor manufacturing equipment ('SME') companies and their competitors in the European Union, Japan, South Korea, and other allied countries."	−2	−1.97
Cerebras BIS-2022-0025	Design	"We respectfully suggest that it is important to support U.S. commercial innovation and leadership by allowing companies to continue to recruit and retain the best and brightest talent, and avoiding overly broad restrictions on the nationalities of available talent."	−1	−1.80
Lenovo BIS-2022-0025	Deployment	"[A]ny definition adopted by BIS should exempt organizations [. . .] with an ultimate parent located in Macau or Country Group D:5, but the day-to-day operations of which are not controlled by an entity located in Macau or Country Group D:5."	−2	−1.51
MathWorks BIS-2022-0025	Deployment	"[W]here BIS must impose restrictions, it should do it even handedly across all distribution models."	+2	0.35
Integra Technologies DOC-2022-0001	Fabrication	"We believe companies that currently operate in China or expand in China that utilize any of these packaging techniques should be subject to clawback provisions."	+2	0.38

Note. Verbatim excerpts. Hand codes from −2 (permanent narrowing) to +2 (more control). Continuous out-of-fold model scores. KLA, relief from US-person controls its allied competitors do not face. Cerebras, deemed exports kept outside the license requirement. Lenovo, exemption from the headquarters test. MathWorks, cloud delivery held to the end-user controls on downloads. Integra, CHIPS docket, expansion clawback extended to advanced packaging.

match the midstream, upstream and downstream stages of value-chain studies (Figure 1). Diversified firms take their largest business, law firms and investors take the client they file for, and small firms with little public information default to the deployment layers as users of AI. Unions and universities have no product, so each of their comments takes the layer of the workers or research it speaks for. Associations, think tanks, individuals and governments have no layer and stay out of H1. In total, 1,657 organization names were coded, and the H1 sample is 1,213 comments from firms, unions and universities in 944 clusters and 36 dockets (1,052 in the deployment layers, 126 in fabrication and 35 in design). A coding relative to each rule's controlled item, done by hand for the semiconductor dockets, is kept for robustness.

3.5 Rule Transitions and Carve-outs

For H2 the unit is a transition from a commented rule to its next version. Seven transitions qualify (Table 4). Three more are dropped, a comment period extension (T4), a rule with no successor (T8) and the AI diffusion framework (T10), which a May 2025 announcement set aside without a rescission in the Federal Register. Each is coded for whether the revision widened, held or narrowed control scope, with corrections that widen in substance counted as widening, and for whether it added any exemption. Within the seven revisions I coded every carve-out, 58 in all, meaning every provision that relieves some party from a control that would otherwise apply. Nine carve-outs soften a control created in the same revision, and they are included. Each is coded for the layer it relieves (one or more of fabrication, design and deployment, or general when it relieves all exporters or the

government) and for whether BIS's own preamble says it granted the relief in response to comments. Each code rests on a quotation from the Federal Register text. I then coded each carve-out against every firm that commented on the rule its revision followed, 33 firm-transitions from 24 firms and 356 pairs of firm and carve-out, leaving out the ten general carve-outs. A carve-out relieves a firm directly when, without it, the revision would bind the firm itself through its own exports, products, plants or US-person staff. It relieves a firm indirectly when the relief reaches the firm's customers or suppliers. Each pair is also coded for whether the firm's own letter asked for that relief. Firm facts come from the letters and from company filings and releases, and 85 close calls are flagged. A wider sample applies the same codes to 119 firms, every firm that commented on a chip rule and every fabrication firm or designer that commented on any docket in the corpus. Each is coded against all 48 carve-outs that relieve a particular layer, 5,712 pairs in all, with 598 close calls flagged.

3.6 Lobbying and Awards

For H3 I build a firm-quarter panel from 2021 to 2025 out of 12,996 LD-2 lobbying disclosures. A filing counts toward a firm's chip and export-control lobbying when its activity text mentions semiconductors, export controls, chips, the CHIPS Act, the Entity List, advanced computing, foundries, wafers, lithography, microelectronics, the Export Administration Regulations, ECCNs or artificial intelligence, or when it lists BIS among the agencies lobbied. CHIPS preliminary and final award dates come from Congressional Research Service (CRS) Report R49031 (Congressional Research Service, 2026). The panel covers 78 firms and associations, with stack layer coded by hand.

4 Who Resists

Table 3 reports comment-level regressions of the scope score on stack layer, with the fabrication layer as the reference. Agency fixed effects are the main specification, since stance means different things across agencies and the CHIPS and AI dockets have their own baselines. Docket fixed effects compare commenters on the same rule. The sample is firms, unions and universities, and the affiliates rule docket is left out because it names no controlled item.

Table 3. Stack layer and stance on control scope

Dependent variable	Scope score			
	(1)	(2)	(3)	(4)
Design or deployment	0.257***		0.253***	0.314***
	(0.097)		(0.092)	(0.091)
	[0.020]		[0.010]	[0.002]
	{0.221}		{0.146}	{0.030}
Deployment		0.280***		
		(0.099)		
		[0.015]		
		{0.197}		
Design		0.057		
		(0.121)		
		[0.646]		
		{0.724}		
Agency fixed effects	Yes	Yes	Yes	No
Argument-type fixed effects	No	No	Yes	No
Docket fixed effects	No	No	No	Yes
Observations	1,212	1,212	1,212	1,211
Comment clusters	943	943	943	943
R^2	0.491	0.492	0.555	0.534
Within R^2	0.008	0.010	0.006	0.009

Note. OLS. Fabrication layer as reference. Standard errors clustered on comment clusters in parentheses. Wild cluster restricted bootstrap p-values by comment cluster in brackets, by docket in braces. Webb weights, 999 draws. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Commenters outside the fabrication layer ask for wider control, or for less narrow-

ing, than fabrication commenters. With agency fixed effects the gap is 0.26 points on the five-point scale (Table 3, column 1). Holding argument type fixed leaves it at 0.25 (column 3), so the gap holds among commenters who make the same kind of argument. With docket fixed effects, which compare firms commenting on the same rule, it is 0.31 (column 4). Restricting the sample to firms gives 0.28 and 0.34 (Appendix Table A3).

Deployers drive the gap, at 0.28 points in column 2. Designers sit close to fabrication firms, but only 35 comments from designers enter the sample and nearly all of them come from outside the chip rules. Chip designers are close to absent from the BIS record. On the chip rules two designers, Cerebras and AMD, filed one comment each, against 24 comments from 14 fabrication firms.

The gap shows up rule by rule where both groups comment. Commenters outside the fabrication layer score 0.66 points above fabrication commenters on the October 2022 rule (9 against 11 comments), 0.63 on the equipment rule, 0.34 on the CHIPS docket (46 against 72) and 0.65 on ITA's 2025 docket on an AI exports program. The one exception is the Section 232 robotics docket, where two fabrication comments sit above 134 others. On the BIS chip rules every group asks for narrowing on average, and fabrication coalitions ask for more of it than deployment coalitions on every rule where both comment (Table 4).

5 Expand and Exempt

Table 4 lists the seven transitions with the mean stance of each coalition on the rule that preceded the revision. Coalitions include firms, unions, universities and associations, which take the layer of their members. On every rule every coalition with at least one member asked for narrower control. Five of the six chip-rule revisions widened scope anyway. The sixth (T7, from the foundry rule to its correction and the April 2026 extension of the designer deadline) held scope flat. The seventh transition, the suspension of the affiliates rule, is the only one that narrowed, and it came by a final rule without comment. Every one of the seven revisions added at least one carve-out.

Table 4. Rule transitions, coalition stance and scope change

				Mean scope score (organizations)		
Transition	From	To	Scope	Design	Fabrication	Deployment
T1	2022-21658	2023-23055, 2023-23049	Wider	−1.86 (1)	−1.40 (10)	−1.38 (7)
T2	2023-23055	2024-07004	Wider	−1.61 (3)	−1.83 (3)	−1.02 (9)
T3	2023-23049	2023-27588, 2024-07004	Wider	–	−1.39 (6)	–
T5	2024-07004	2024-28270	Wider	–	−1.10 (4)	−0.74 (2)
T6	2024-28270	2025-00711	Wider	–	−1.78 (4)	−0.30 (2)
T7	2025-00711	2025-02655, 2026-06851	Flat	−1.97 (1)	−1.44 (3)	−1.42 (2)
T9	2025-19001	2025-19846	Suspended	−1.08 (2)	−1.38 (3)	−1.24 (7)

Note. Federal Register document numbers. Mean fine-tuned scope score by layer. Number of organizations in parentheses.

The pattern contradicts the view that firms set the direction of the controls. No coalition asked for wider control on any of these rules, and the coalitions differed only in how hard they pushed for narrowing.

Table 5 shows where the carve-outs landed. 35 of the 58 relieve fabrication firms, 29 of them fabrication firms alone. Fourteen relieve designers and 13 relieve deployers, mostly in

exceptions shared with another layer. Ten are general, such as savings clauses for shipments en route and exceptions for US government use. BIS says that 21 of the 58 respond to comments, and 15 of those 21 reach fabrication firms. In the same rules, fabrication-layer items account for about a third of the controlled ECCN provisions (110 of 329). Relief went to fabrication firms at roughly twice their share of what was controlled.

Table 5. Carve-outs in seven BIS revisions by the layer of the stack they relieve

	All carve-outs	Granted at commenters' request	Reopens flows to China
Reaches fabrication firms	35	15	18
fabrication firms alone	29	13	18
Reaches designers	14	4	4
Reaches deployers	13	3	4
General (all exporters or the government)	10	2	1
Total	58	21	24

Note. Carve-outs can reach several layers. Layer rows overlap.

The carve-outs also differ in what they let through. I coded each one into three groups. The first lets items, technology or US-person services reach end users in China without a license. The second relieves operations outside China, allied staff or firms' own supply chains, and the third settles timing or administration. Allied firms' reexports to China fall in the first group, and items sent into China for US or allied firms' own supply chains fall in the second. Twelve of these codes are close calls, six on each side of the line. 24 of the 58 reopen flows to China in this sense, and 18 of those 24 reach fabrication firms (Table 5). All 8 requested carve-outs that reopen flows to China reach fabrication firms. About half of the relief for fabrication firms reopened flows to China (18 of 35), against 4 of 14 carve-outs for designers and 4 of 13 for deployers. Resolving every close call one way or the other leaves fabrication firms with between 73 and 83 percent of the carve-outs that reopen flows to China.

Table 6 regresses the number of carve-outs that directly relieve each commenting firm on the firm's scope score on the rule it commented on, with transition fixed effects. HORIBA STEC, the only firm commenting on T9, drops with its transition, and the count models also drop any transition in which no firm received the outcome. Firms that asked for more narrowing received more direct relief. In the Poisson model in column 2, a one-point move toward narrowing on the five-point scale goes with about twice as many carve-outs relieving the firm. The negative binomial model gives about 1.9 times as many (column 3) and the OLS model about one more carve-out per point (column 1). The gap between the average fabrication firm and the average deployer, 0.74 points on the scale, corresponds to about 1.6 times as many carve-outs. Counting only codes without close calls, a one-point move goes with about 4.3 times as many (column 4).

Table 6. Scope score and carve-outs relieving each commenting firm

Dependent variable	Number of carve-outs						
Outcome	Direct	Direct	Direct	Direct, clear	Own request	Direct	Direct
Estimator	OLS	Poisson	Neg. bin.	Poisson	Poisson	Poisson	Poisson
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Scope score	-1.004^{*}	-0.676^{*}	-0.634^{**}	-1.463^{***}	-0.600	-0.645^{*}	-0.491
	(0.543)	(0.377)	(0.307)	(0.554)	(0.876)	(0.387)	(0.378)
	[0.070]						
Fabrication layer							0.625
							(0.545)
Transition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	32	32	32	30	20	29	32

Note. Unit, a firm commenting on the rule a transition starts from. Standard errors clustered on firm in parentheses. Wild cluster restricted bootstrap p-value in brackets. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Firms with many product lines collected the most relief. Applied Materials, which makes deposition, etch, implant, inspection and packaging tools and supports fabs in China, was relieved directly by 13 of the 23 carve-outs in the October 2023 package that relieve particular layers, and Intel, which designed controlled chips and ran a memory fab in Dalian,

by 10. Both were among the hardest pushers on the 2022 rule, with scores of −1.98 and −1.86. Specialists collected little however hard they pushed. Momentive, a maker of fused quartz, scored −1.99 and was relieved by one carve-out, STMicroelectronics by one and Seagate by none. A rule reaches a broad firm at more points, so the breadth of a firm's exposure accounts for part of the link between resistance and relief.

The scope score does not predict how many carve-outs granted what a firm's own letter asked for (column 5). Relief tracked how hard a firm resisted and how much it stood to lose more closely than what it asked for, which fits a state that chooses the form of payment for the firms it needs to manage. The request measure is also incomplete. BIS rarely names commenters, the business-confidential versions of the Applied, KLA, Lam and Samsung letters are not public, and associations such as SIA and SEMI filed many requests for their members.

The estimate rests on 32 firm-transitions from 23 firms and leans on the largest firms. It keeps its sign when any single firm is dropped, though dropping Applied, Intel or KLA weakens it. Without transition fixed effects the direct-relief estimates are larger in all three models. The letters on the December 2024 rule are dated March 14, 2025, after the January 2025 revision they are paired with, so their scores cannot have shaped its carve-outs. Without that transition, a one-point move goes with about 1.9 times as many carve-outs in the Poisson model (column 6). Stance and layer are hard to separate. With a fabrication-layer indicator added, neither term predicts direct relief (column 7), although the scope score keeps its link to relief when only clear calls are counted.

The test above draws only on the firms that wrote on the rules the revisions followed. Most firms in the corpus commented elsewhere, on the CHIPS incentives program, the Section 232 investigations and the AI requests for information. The wider sample adds the fabrication firms and designers among them. Each firm's stance is its mean scope score net of the mean on each docket it commented on, so a firm that wrote only on the CHIPS docket is compared with other CHIPS commenters. The outcome counts the carve-outs across all seven revisions that relieve the firm directly.

Table 7 reports the results. Firms that asked for more narrowing were relieved by more carve-outs. In the Poisson model in column 2, a one-point move toward narrowing goes with about 1.9 times as many carve-outs. The OLS model gives about 1.6 more carve-outs per point (column 1), the negative binomial model about 2.7 times as many (column 3), and the Poisson model on clear calls about 2.4 times as many (column 4). This sample separates stance from layer, which the smaller test could not do. With a fabrication-layer indicator added, the scope score keeps its size, about 1.9 times as many carve-outs per point, and the indicator is close to zero (column 5). Among the 94 firms that never commented on a chip rule, a one-point move goes with about 1.7 times as many (column 6). The estimate also holds without the two law firms that filed for unnamed clients and when any single firm is dropped.

The wider sample gives up the timing of the transition test. Stance comes from comments filed at any point between 2022 and 2026, and for most of the 94 firms it was measured on CHIPS conditions, tariffs or AI policy, with each docket's mean removed.

Table 7. Scope score and carve-outs relieving each firm in the wider sample

Dependent variable	Number of carve-outs relieving the firm					
Outcome	Direct	Direct	Direct	Direct, clear	Direct	Direct
Estimator	OLS	Poisson	Neg. bin.	Poisson	Poisson	Poisson
Sample	All	All	All	All	All	Off chip dockets
	(1)	(2)	(3)	(4)	(5)	(6)
Scope score	−1.619**	−0.635***	−0.982***	−0.877***	−0.641***	−0.538**
	(0.665)	(0.234)	(0.331)	(0.310)	(0.231)	(0.273)
Fabrication layer					−0.059	
					(0.354)	
Observations	119	119	119	119	119	94

Note. Unit, a firm. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The carve-outs that BIS granted on request show the mechanism. In the October 2023 equipment rule BIS narrowed its end-use control on equipment for Chinese toolmakers from any item subject to the EAR to items on the Commerce Control List, writing that "BIS agrees that this provision is overbroad." It exempted back-end assembly, test and packaging from the definition of production, excluded masks and reticles, gave US and allied equipment makers a temporary general license to keep using suppliers in China, and excluded US-person employees of allied fabs and equipment firms from the US-person controls. It dropped a control on cobalt deposition tools, saying that "based on feedback from industry, including from this commenter, BIS agrees" that the control was poorly tailored. In one case the rule text tracks a comment closely. An anonymous commenter on the 2022 rule drafted replacement text for the control on atomic layer deposition tools, and BIS adopted nearly the same four-condition structure in 2023 while setting a different temperature threshold. The January 2025 foundry rule, which widened the presumption of control over chips made by foundries, was issued "in response to requests from the public" for due diligence procedures, and its approved designer and approved packager lists let foundries and packaging firms

certify their way out of the presumption.

In each case the rule that granted relief to fabrication firms also extended the controls over the same firms' products and plants.

6 Paying the Resisters

CHIPS money went to the coalition that resisted most. Of $33.5 billion in incentive awards that reached preliminary or final terms, $32.8 billion went to 26 fabrication firms, the foundries, memory makers, integrated manufacturers, equipment makers and materials suppliers.[2] No designer received one.

6.1 From Lobbying to Award

H3a asks whether lobbying predicts receiving preliminary terms. Table 8 reports a complementary log-log hazard of preliminary terms by quarter, with the risk set running from the first to the last quarter in which any firm received terms, calendar-quarter effects and controls for size and stack layer. Associations are dropped. Filings and spending enter as the log of one plus the value. The sample has 68 firms, 357 firm-quarters and 26 events.

No lobbying measure predicts preliminary terms, although every estimate points in the expected direction (Table 8). With 26 events the test has little power. An event study

[2]Deployers among the recipients, by main business, are HP, Infinera, Rocket Lab, Corning, BAE Systems and Bosch, about $0.6 billion together. Arizona State University and Activate account for the rest.

Table 8. Lobbying and the hazard of receiving CHIPS preliminary terms

Dependent variable	Preliminary terms			
	(1)	(2)	(3)	(4)
Filings, previous quarter (log)	0.318			
	(0.302)			
Filings, previous four quarters (log)		0.389		
		(0.259)		
Filings, cumulative to date (log)			0.362	
			(0.299)	
Spend, previous four quarters (log)				0.093
				(0.063)
Firm size and stack layer	Yes	Yes	Yes	Yes
Calendar-quarter fixed effects	Yes	Yes	Yes	Yes
Firm-quarters	357	357	357	357
Preliminary terms (events)	26	26	26	26

Note. Complementary log-log discrete-time hazard. One model per lobbying measure. Log hazard ratios. Standard errors clustered by firm in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

of the same panel, comparing firms that received preliminary terms with firms that never did, shows lobbying rising before terms were announced, with about 13 to 15 percent more filings two quarters before (Figure 2). The award appears to be partly a product of lobbying, which is why the next step compares only firms that had already received terms.

6.2 From Award to Demobilization

H3b compares the 16 firms in the panel whose awards were finalized with the 10 that received preliminary terms and never finalized. Holding preliminary terms fixed holds selection into the program fixed, so the pre-period coefficients carry their usual meaning as a test of parallel trends. Figure 3 plots event-time estimates from four estimators (Dube et al., 2025; Sun & Abraham, 2021; Wooldridge, 2023).

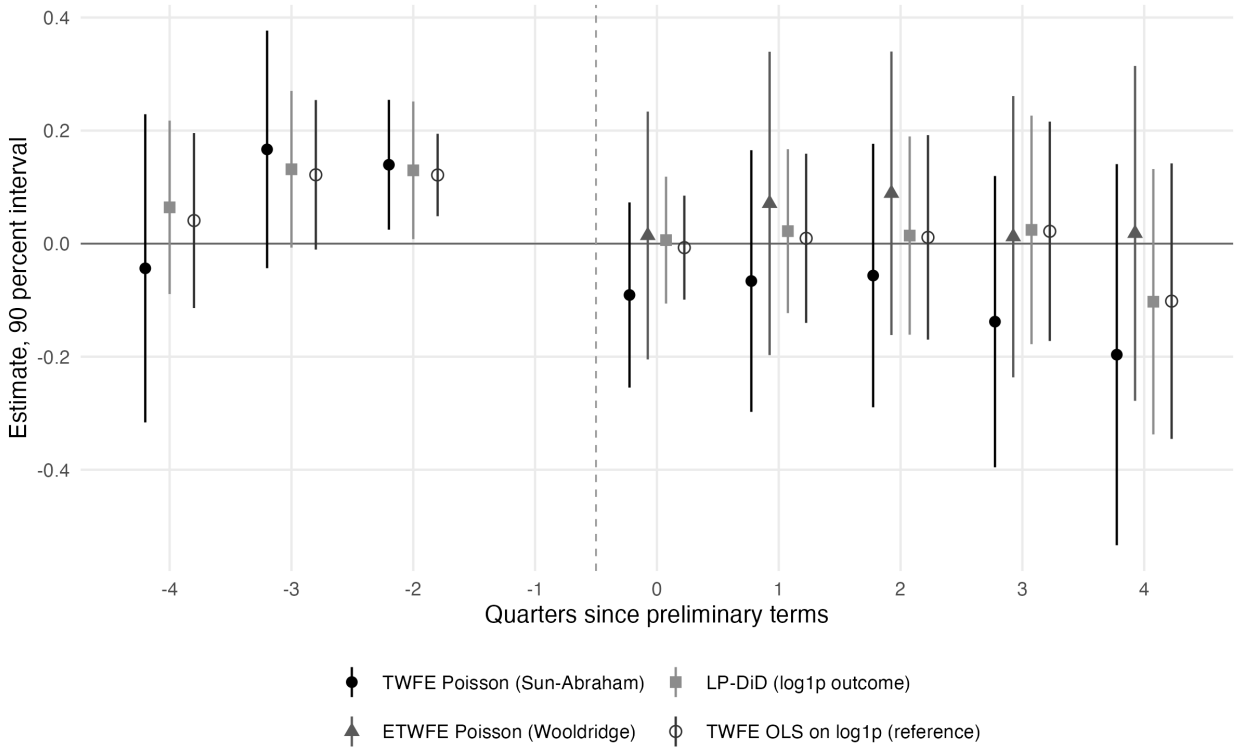


Figure 2. Chip and export-control lobbying around CHIPS preliminary terms, firms that received terms against firms that never did. 77 firms, 25 with terms. Points are event-time estimates with 90 percent intervals. Event −1 is the reference. Pre-period estimates describe the H3a mechanism and are not a test of parallel trends.

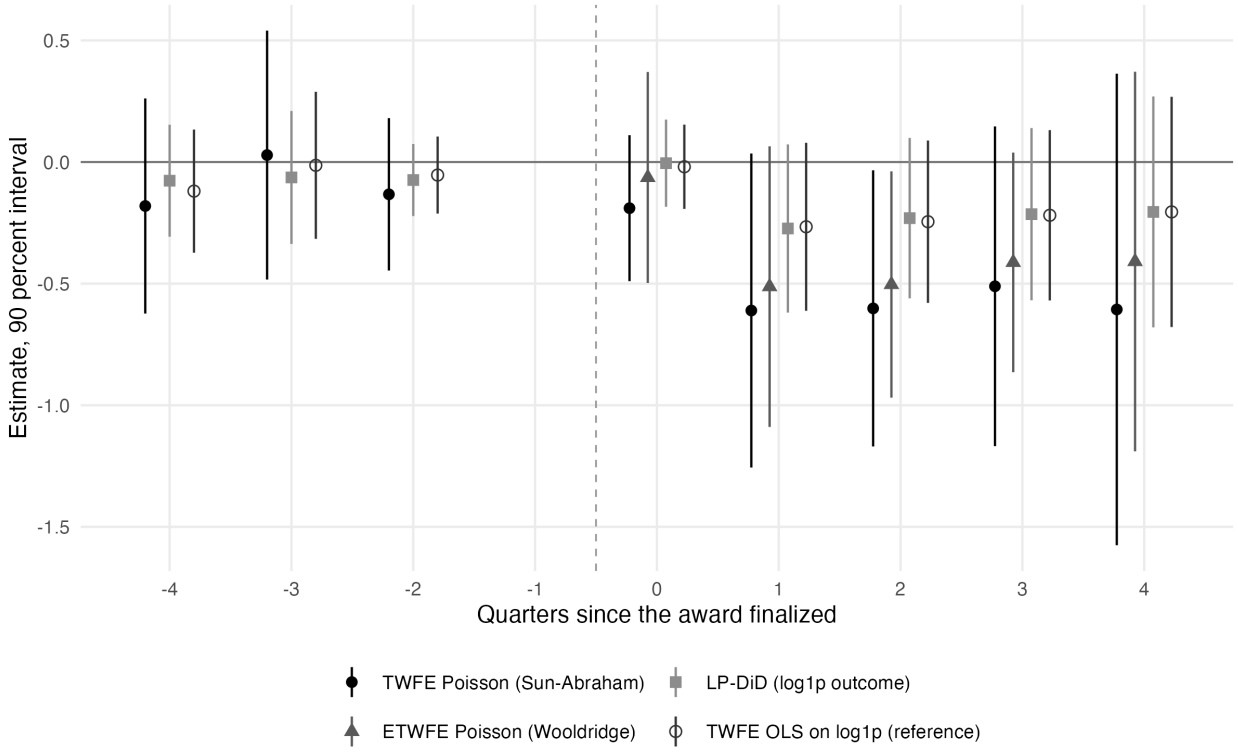


Figure 3. Chip and export-control lobbying after CHIPS award finalization, finalized firms against firms with preliminary terms only. Outcome is LD-2 filings per firm-quarter. Points are event-time estimates with 90 percent intervals. 26 firms, 16 finalized. Event -1 is the reference.

Before finalization the two groups move together, and every pre-period estimate is small. After finalization, finalized firms file less. Two quarters after the award, filings fall by about 45 percent in the Sun and Abraham Poisson model and by about 40 percent in the Wooldridge Poisson model (Figure 3). The log-linear estimators show no clear decline. Poisson weights by expected counts, and Micron, Intel and Samsung account for much of the filing volume, so the decline is concentrated among the largest filers. The design is thin. Final awards fall in two quarters, 2024Q4 and 2025Q1, and the panel ends in 2025Q4.

6.3 Timing

Figure 4 lays the award dates over the rule calendar. 20 of the 21 final awards came between November 15, 2024 and January 17, 2025. The December 2, 2024 rule, which added equipment controls and foreign direct product rules aimed at equipment makers, took effect within three weeks after TSMC, GlobalFoundries and Intel finalized their awards, and Micron, SK hynix, Samsung and Texas Instruments finalized theirs in the three weeks that followed. The January 16, 2025 foundry rule took effect the same week that Applied Materials, Corning, Edwards Vacuum and Infinera finalized theirs.

7 Statecraft or Private Interest

US chip export controls are security statecraft that private interest hollows out from within. The state set the direction of the controls. Every coalition that commented

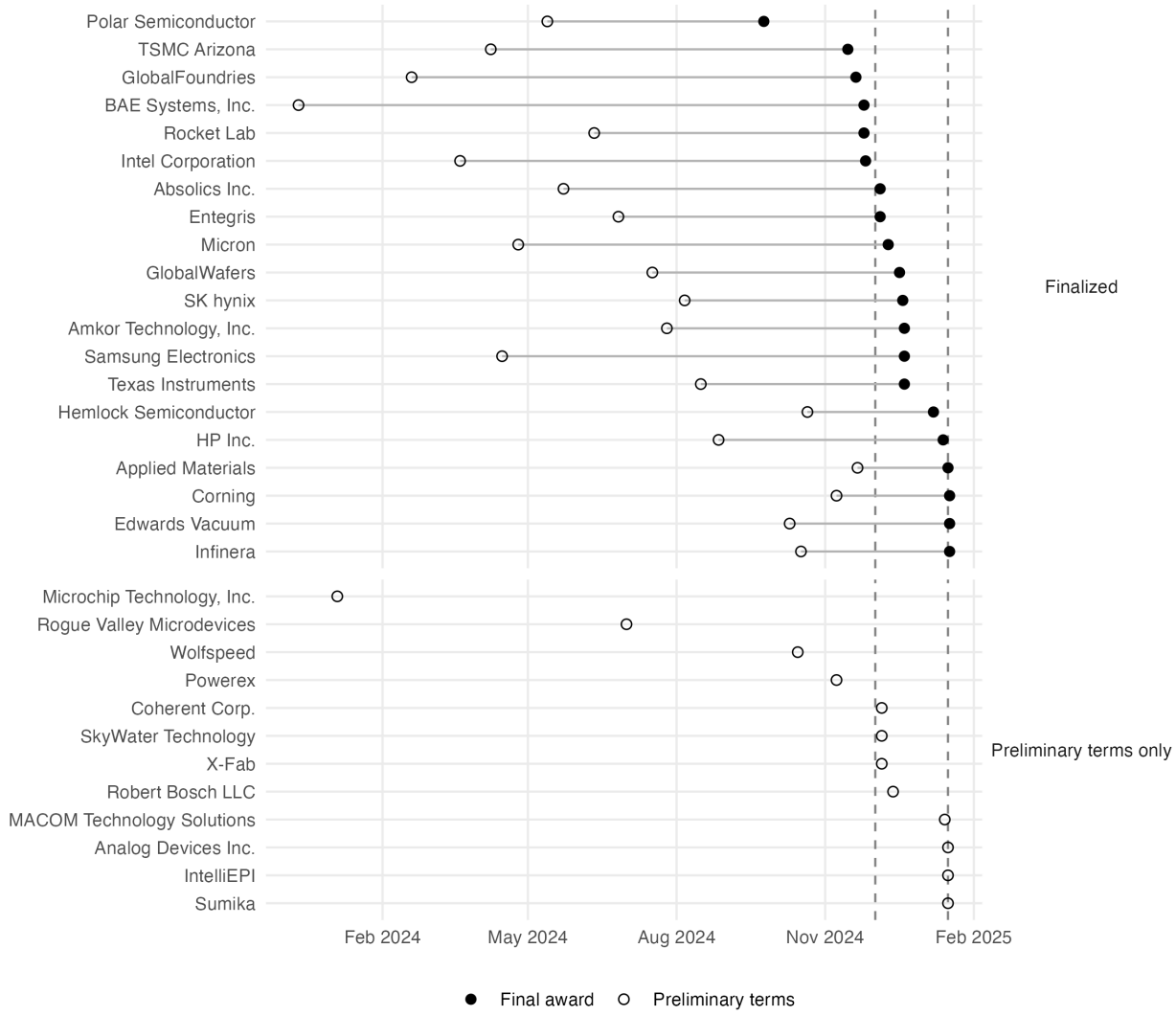


Figure 4. CHIPS preliminary terms and final awards by firm. Dashed lines mark the December 2, 2024 rule on equipment, foreign direct product rules and high bandwidth memory, and the January 16, 2025 foundry due diligence rule. Dates from CRS Report R49031.

asked for less control on every chip rule, including the deployers that the commercial reading expects to gain from wider controls, and BIS widened scope in five of six revisions anyway. The firms that resisted hardest decided where the wider controls would bite. Relief went to fabrication firms at roughly twice their share of the controlled provisions, and 15 of the 21 carve-outs that BIS granted at commenters' request reached them. Among the firms that commented, those that pushed harder for narrowing were relieved by more carve-outs, and the relief followed what each firm stood to lose more closely than the specific requests in its letter. The same pattern appears across 119 firms in the wider corpus, and it holds among firms in the same layer of the stack.

Exemptions for fabrication firms carry a security cost because those firms hold a chokepoint. A state's leverage in an interdependent network rests on the hubs it controls (Farrell & Newman, 2019), and advanced equipment is one of the US hubs in chipmaking. In a stack the cost of relief at the base carries upward. The tools that reach Chinese fabs set which chips China can make, and those chips set how much computing Chinese cloud providers and model developers have to build on. 18 of the 24 carve-outs that reopened flows to China went to fabrication firms, including all 8 that BIS granted at commenters' request. The October 2023 equipment rule narrowed the end-use control on Chinese toolmakers from any item subject to the EAR to items on the Commerce Control List, so uncontrolled US inputs could again reach Chinese firms that build chipmaking equipment and compete with US toolmakers. The same rule dropped a control on cobalt deposition tools. Outside the revisions coded here, Validated End-User authorizations let suppliers ship controlled items to Samsung's and SK hynix's memory plants in China without licenses from October 2023

until the end of 2025 (88 FR 71478). Most of the remaining relief for fabrication firms eased allied operations or firms' own supply chains. The carve-outs that reopened flows to China gave up part of the leverage the controls were built on, and most of them went to the firms that hold that leverage.

Chip designers show the same pattern outside the rulemaking record. They lose most from controls on chip sales, filed almost no comments and could not be paid through manufacturing incentives. They had already built chips to fall just below the 2022 thresholds, which BIS redefined in October 2023 to capture them. Their exemption came from the White House. In August 2025 the President said he had agreed to approve Nvidia's sales of its H20 chip to China in return for 15 percent of the revenue (Trump, 2025a). On December 8, 2025 he announced that Nvidia could ship its H200 to approved customers in China with 25 percent paid to the United States (Trump, 2025b), and in January 2026 BIS moved H200 license review to case-by-case by a final rule issued without comment (Bureau of Industry and Security, 2026). Here the exempted item was the advanced AI chip itself, the item the October 2022 rule was first written to keep from China. These decisions came under a new administration and alongside a trade arrangement with China (The White House, 2025), which may explain their timing. They still followed the same line, with relief going to the firm whose sales the controls had cut.

The CHIPS money worked differently from the carve-outs. Final awards bound recipients to the ten-year Expansion Clawback, so the payment carried conditions that served the security aim, and the paid firms lobbied less on chips and export controls once their awards were secure. The state paid its strongest resisters twice, in the rule text and in cash,

and the cash came with security conditions attached.

Expand and exempt means the scope written into the rules overstates how much the controls restrict China, because the exemptions sit where US leverage is greatest. The security label also shelters the bargain. Calling an issue a security matter lifts it out of normal politics (Buzan et al., 1998), and the exemptions were negotiated inside that frame, first in BIS's responses to comments and later in announcements by the President.

8 Conclusion and Future Work

US chip export controls are security statecraft that private interest hollows out from within. They widened while every organized coalition in the record asked for less, and the firms that resisted hardest decided where the wider controls would bite. Of the 24 carve-outs that reopened flows to China, 18 went to fabrication firms, including all 8 that BIS granted at commenters' request. These firms make the equipment and materials and run the fabs that give the United States its hold over Chinese chipmaking, so the relief they won came out of the leverage the controls were meant to use.

Counts of controlled items and listed entities therefore overstate how far the United States has restricted China, because the exemptions are concentrated in chipmaking equipment and services, where US leverage is greatest. Assessments of the controls need to read the carve-outs alongside the expansions. The form of payment matters as well. CHIPS awards came with a ten-year bar on expanding capacity in China, so the cash bought quiet

on terms that served the security aim, while the carve-outs bought quiet by reopening flows to China. If the state wants wider controls to keep their bite, paying the losers in cash with conditions costs less security than paying them in exemptions.

For the study of economic statecraft, the case shows a limit on weaponized interdependence that comes from home (Farrell & Newman, 2019). The firms that own a chokepoint give the state its leverage and lose the most when the state uses it, so their resistance is aimed at the source of that leverage. The more a state relies on chokepoints held by its own firms, the more its controls depend on how it pays those firms.

Chip designers are the gap in this record. They lose most from controls on chip sales, are nearly absent from the comment record and cannot be paid through manufacturing incentives. Their exemptions came from the White House after 2025, in sales approvals for Nvidia's H20 and H200 that traded access to China for a share of revenue. Earnings call transcripts for US-listed firms across the stack would bring designers into the H1 comparison, give H3b a stance measure for firms with preliminary terms only, and address selection into commenting (Hassan et al., 2019, 2024).

Validation of the scorer is strong on the BIS dockets, weaker on the CHIPS docket and absent on the AI dockets that supply most of the H1 sample. The lobbying measure counts filings that mention chips, export controls or the CHIPS Act, so part of the post-award decline may reflect firms that stopped lobbying for an award they had won. A count restricted to export-control activity would separate the two. The H3b comparison rests on 26 firms and two award quarters. The carve-out coding counts provisions and treats each

one as a unit regardless of the trade it covers, and twelve of the flow codes are close calls. The firm-level codes cover 33 firm-transitions, the estimates for all direct relief lean on the largest firms, and the letters on the December 2024 rule postdate the revision they are paired with. The wider test across 119 firms gives up the order of comment and revision, and most of its stance scores come from dockets other than the chip rules. Weighting carve-outs by trade would show how much of the controls' reach the exemptions gave back.

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A. Appendix

A.1 Scorer Validation

Table A1 reports out-of-fold agreement between the scorer and the hand codes. Scope F1 is weighted and cuts the continuous score at the shares of the hand-coded labels. The base model is the same NLI model before fine-tuning. The pooled Spearman correlation is inflated by the level gap between BIS and CHIPS letters. For argument type, recall and precision are 0.89 and 0.85 for fabrication, 0.64 and 0.58 for design inputs and 0.52 and 0.54 for deployment.

Table A1. Out-of-fold validation of the fine-tuned scorer against hand codes

	Letters	Spearman	Scope F1	Argument type
All hand-coded letters	213	0.78	0.61	0.78
BIS dockets	182	0.73	0.60	0.81
CHIPS docket (DOC-2022-0001)	31	0.49	0.39	0.61
BIS dockets, base model	182	0.64	0.53	0.50

Note. Out-of-fold scores. Weighted scope F1. Exact agreement on argument type.

A.2 Long-Context Scorer

The scorer reads each comment in windows of 1,800 characters, so its score for a long letter is built from scores for its parts. To check whether reading a letter whole changes the measure, I refit the scorer with ModernBERT-large-zeroshot-v2.0, an NLI model built on ModernBERT, which reads up to 8,192 tokens at once (Warner et al., 2025). Each comment enters whole as a single premise, in training and in scoring. The statements, labels, class

balance, epochs and learning rate are those of the main scorer, and the folds are the same, so both scorers hold out the same letters. Nineteen of the 213 hand-coded letters run past 8,192 tokens and are cut at the end.

Table A2 compares the two scorers against the hand codes. The long-context scorer agrees less with the hand codes on all three measures, both across all letters and on the 194 letters it reads in full. On the 19 letters that exceed its window it does better on stance and worse on argument type. The stance scores of the two scorers correlate at 0.78 (Spearman). Reading whole letters does not bring the scores closer to the hand codes, so the main analysis keeps the windowed scorer.

Table A2. Windowed and long-context scorers against hand codes

		Spearman		Scope F1		Argument type	
	Letters	Windowed	Long	Windowed	Long	Windowed	Long
All hand-coded letters	213	0.78	0.73	0.61	0.53	0.78	0.66
Within 8,192 tokens	194	0.77	0.71	0.59	0.51	0.78	0.68
Past 8,192 tokens	19	0.84	0.86	0.37	0.47	0.84	0.53

Note. Out-of-fold scores. Windowed, DeBERTa-v3 on 1,800-character windows. Long, ModernBERT on whole comments. Weighted scope F1. Exact agreement on argument type.

A.3 Firms Only

Table A3. Stack layer and stance on control scope, firms only

Dependent variable	Scope score			
	(1)	(2)	(3)	(4)
Design or deployment	0.285***		0.285***	0.345***
	(0.097)		(0.088)	(0.091)
	[0.010]		[0.004]	[< 0.001]
	{0.209}		{0.175}	{0.032}
Deployment		0.303***		
		(0.098)		
		[0.005]		
		{0.202}		
Design		0.110		
		(0.122)		
		[0.372]		
		{0.578}		
Agency fixed effects	Yes	Yes	Yes	No
Argument-type fixed effects	No	No	Yes	No
Docket fixed effects	No	No	No	Yes
Observations	1,152	1,152	1,152	1,151
Comment clusters	900	900	900	900
R^2	0.498	0.499	0.566	0.542
Within R^2	0.010	0.012	0.008	0.012

Note. Specifications as in Table 3, firms only. Standard errors clustered on comment clusters in parentheses. Wild cluster restricted bootstrap p-values by comment cluster in brackets, by docket in braces. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.4 Codebook Summary

Scope is coded on one five-point scale. -2 requires a request for permanent narrowing of something the rule already covers, such as an exclusion, a carve-out, a raised threshold, a rollback or withdrawal, and drafted replacement text that does this counts. Rejecting the premise of controlling at all is also -2. -1 covers transitional relief, clarification requests that lean narrow, resistance to controls that do not yet exist and conditional narrowing. 0

means no net position, including clarification requests with no preferred answer. +1 endorses the rule as written or proposes drafting that leans wider without pressing for more capture. +2 asks that more be captured. Scope records direction only. A firm that asks for controls on its rivals' delivery channel is coded as widening. When asks conflict, the main argument sets the sign and the effect on the target country decides close cases.

Argument type records where the commenter argues from, regardless of side. Clarification letters are coded by the locus of the activity being defended, and narrowing letters by the locus of their narrowing asks, with the letter's emphasis taking precedence over a count of routine asks. Compliance process is reserved for letters whose grievance is the licensing machinery itself, and national security for letters whose end is security. On the CHIPS docket, scope is stance on the restrictions and conditions attached to CHIPS money, from asking to remove a statutory condition (-2) to asking to add or extend one ($+2$).

Firm relief is coded for each pair of a commenting firm and a carve-out in its transition that relieves a particular layer. Direct relief means that, in the revision without that carve-out, a license requirement or restriction would fall on the firm's own exports, products, plants, affiliates, US-person staff or transactions. Indirect relief reaches the firm's customers, suppliers or partners. A carve-out whose relief another carve-out in the same revision already gives is coded as no relief for that firm. The request code records whether the firm's own letter asked for the relief or for something the carve-out substantially grants, and a general plea for narrower controls does not count. Close calls are flagged.